

RAG Service Architecture

Internal technical overview - bilingual training sample

1. Offline ingestion

Documents are parsed, OCR is applied only to text-poor PDF pages, and content is split into metadata-preserving chunks.

2. Online query path

The API normalizes the current question, retrieves candidate chunks, optionally reranks them, checks confidence, and sends grounded context to the generation model.

3. Reliability boundaries

Provider timeouts return controlled errors. Low-confidence and out-of-scope questions return guided refusals. Every request carries a request ID and trace ID.

Component Responsibilities

API: validation, session handling, response contract, and request correlation.

Retriever: vector-only or hybrid search selected by configuration.

Reranker: optional relevance refinement of the top candidate chunks.

Generator: context-grounded answer, citation construction, and refusal behavior.

Observability: structured events, stage latency, tokens, cache, safety, and errors.

All components emit structured events so failures can be diagnosed by request and stage.